



Integrated water quality monitoring system with pH, free chlorine, and temperature sensors



Yiheng Qin^{a,b}, Arif U. Alam^{a,b}, Si Pan^c, Matiar M.R. Howlader^a, Raja Ghosh^c, Nan-Xing Hu^{b,*}, Hao Jin^{a,d}, Shurong Dong^d, Chih-Hung Chen^a, M. Jamal Deen^{a,*,1}

^a Department of Electrical and Computer Engineering, McMaster University, 1280 Main Street West, Hamilton, ON, L8S 4K1, Canada

^b Advanced Materials Laboratory, Xerox Research Centre of Canada, 2660 Speakman Drive, Mississauga, ON, L5 K 2L1, Canada

^c Department of Chemical Engineering, McMaster University, 1280 Main Street West, Hamilton, ON, L8S 4L7, Canada

^d College of Information Science and Electronic Engineering, Zhejiang University, 38 Zheda Road, Hangzhou, 310027, PR China

ARTICLE INFO

Article history:

Received 10 May 2017

Received in revised form 29 June 2017

Accepted 26 July 2017

Available online 16 August 2017

Keywords:

Inkjet printing

Integration

pH

Free chlorine

Temperature

Water quality monitoring

Sensor

ABSTRACT

Accurate, efficient, inexpensive, and multi-parameter monitoring of water quality parameters is critical for continued water safety from developed urban regions to resource-limited or sparsely populated areas. This study describes an integrated sensing system with solution-processed pH, free chlorine, and temperature sensors on a common glass substrate. The pH and temperature sensors are fabricated by low-cost inkjet printing of palladium/palladium oxide and silver. The potentiometric pH sensor has a high sensitivity of 60.6 mV/pH and a fast response of 15 s. The Wheatstone-bridge-based temperature sensor shows an immediate response of 3.35 mV/°C towards temperature change. The free chlorine sensor is based on an electrochemically modified pencil lead, which exhibits a stable and reproducible sensitivity of 342 nA/ppm for hypochlorous acid. Such a free chlorine sensor is potentiostat-free and calibration-free, so it is easy-to-use. The three sensors are connected to a field-programmable gate array board for data collection, analysis and display, with real-time pH and temperature compensation for free chlorine sensing. The developed sensing system is user-friendly, cost-effective, and can monitor water samples in real-time with an accuracy of >82%. This platform enables water quality monitoring by nonprofessionals in a simple manner.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

The monitoring of water quality is extremely important for maintaining the safety of water resources used for various purposes such as drinking, recreation or food processing [1–5]. Water quality is determined by interdependent chemical, physical, and microbial factors, including but not limited to pH, free chlorine, dissolved oxygen, organic carbon, turbidity, conductivity, temperature, and microorganisms [6]. Many conventional water quality monitoring technologies lack integration, are labor-intensive, time-consuming, and expensive [4]. Typically, in many systems, only one parameter such as free chlorine [7] or pH [8] is measured at a time [9]. These technical and economic limitations pose imminent challenges in

maintaining high water quality in both heavily-populated cities (large water consumption and fast water quality degradation) and in remote areas (resource-limited) [10]. Therefore, low-cost, highly sensitive, accurate, reliable, easy-to-use, and integrated sensing systems are needed for the regular/continuous monitoring of multiple water quality parameters in multiple locations to ensure continued water safety.

Free chlorine (the sum of hypochlorous acid and hypochlorite ion) is widely used for water disinfection, and the free chlorine concentration should be accurately monitored for the water safety, especially for drinking, recreational, and food processing water [11]. The World Health Organization recommends the usage of 2–3 ppm free chlorine in the drinking water [1]. The free chlorine concentration in a swimming pool should be maintained between 1.5–2 ppm [12]. For food processing, the water should contain 50–200 ppm free chlorine [13]. The accurate monitoring of free chlorine concentration is challenging because it is correlated with the temperature and pH of the water sample [14]. For example, the dissociation constant of hypochlorous acid (HOCl) in water is a function of temperature. The pH of the water sample determines

* Corresponding authors.

E-mail addresses: nan-xing.hu@xerox.com (N.-X. Hu), jamal@mcmaster.ca (M.J. Deen).

¹ The development of the pH and temperature sensors was carried out through the collaboration between a and b, the development of the free chlorine sensor was carried out through the collaboration between a and c.

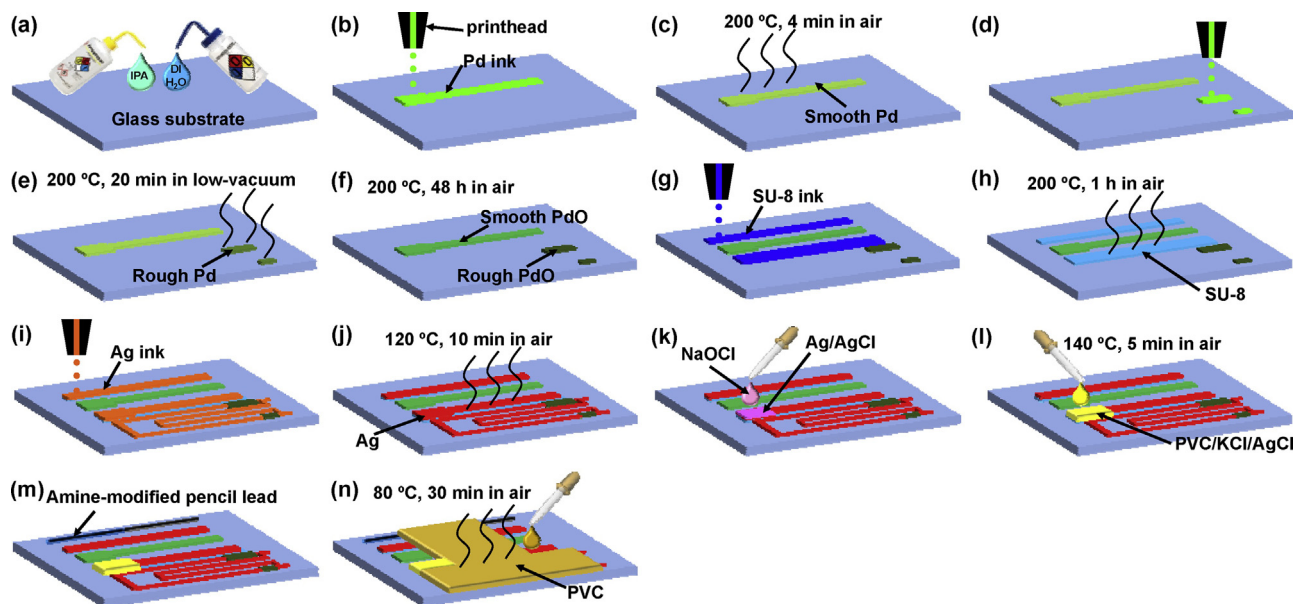


Fig. 1. Fabrication process of an integrated pH, free chlorine and temperature sensor.

the concentration ratio between HOCl and hypochlorite ion (OCl^-) at a certain temperature. In addition, the response of a pH sensor is dependent on the temperature [5]. Thus, pH and temperature sensors are required and should be integrated into a free chlorine sensing system.

The standard method to measure the free chlorine concentration is to use a *N,N'*-diethyl-*p*-phenylenediamine (DPD)-based absorptiometry method [15]. However, this optical method is difficult to use, of high-cost, difficult to perform continuous monitoring, and the free chlorine reagent contains chemicals that are harmful to humans, such as the DPD compound and ethylenedinitrilotetraacetic acid disodium salt dihydrate. Therefore, amperometric electrodes are utilized for the online monitoring of free chlorine [16,17]. One of the challenges in using the existing amperometric sensors is the requirement of frequent calibration due to the poor stability of the sensors. Chemoresistor-based free chlorine sensors are not suitable for continuous monitoring because the sensing materials need to be dried/activated after each/several measurement(s) [13,18]. For both optical and electrochemical free chlorine sensing systems, pH sensors and temperature sensors are usually not integrated on a common platform (at most connecting a discrete pH/temperature sensor with a cable).

For pH sensing, conventional glass electrodes are bulky, fragile and costly. These electrodes are also difficult-to-use due to the need for frequent calibration and the requirement of complicated electronic circuits for signal conditioning (amplification, filtering, etc.) [5]. To improve conventional pH meters, many studies focused on developing microfabricated pH sensors with smaller dimensions, higher stability, simpler operation, and lower cost [4,5,19–21]. Also, there are many studies on micro-scale temperature sensors [22–26]. One of the most commonly used temperature sensors is a thermistor made from metals [22], polymers [23], ceramics [24], carbon nanomaterials [25], or their composites [26]. However, the technologies to fabricate low-cost, integrated pH and temperature sensors that are compatible with other water quality sensors are still to be explored.

Recently developed integrated sensing systems have already shown advantages in human perspiration analysis [27,28] and diabetes monitoring/therapy [25]. For example, the integrated sensor arrays had a high spatial and temporal sensing resolution, and the integrated electronics could perform simultaneous on-site

signal processing. Thus, the integrated systems were compatible with multiplexed analysis. For water quality monitoring, integrated sensors were developed for measuring dissolved oxygen, conductivity, temperature, pH, bacteria (*Escherichia coli*), and cell nutrients/metabolites [29–33]. However, an integrated free chlorine monitoring system is not yet developed.

In this study, we report on an integrated sensing system which could simultaneously measure the pH, free chlorine concentration, and temperature of a water sample. The potentiometric pH sensor was fabricated by inkjet printing of palladium/palladium oxide (Pd/PdO) as the sensing electrode and silver/silver chloride/potassium chloride (Ag/AgCl/KCl) as the reference electrode. The printed pH sensor had a high sensitivity and did not require signal conditioning for data analysis. A Wheatstone-bridge-based temperature sensor was inkjet-printed together with the pH sensor using the same Pd and Ag ink. The temperature sensor showed a higher sensitivity compared to a single thermistor. The free chlorine sensor is a potentiostat-free amperometric sensor, consisting of an electrochemically modified pencil lead and an inkjet-printed Ag electrode. Such a free chlorine sensor was calibration-free due to its high reproducibility and stability. The three solution-processed sensors were on a common glass substrate, and they were connected to a programmed field-programmable gate array (FPGA) board for on-site monitoring of real water samples. A high measurement accuracy was obtained. This indicated that the integrated sensing system has the potential to be used as a rapid, user-friendly, cost-effective, and widely applicable method for water quality monitoring.

2. Materials and methods

2.1. Chemicals and reagents

Isopropanol (IPA, 8600-1), acetic acid (HOAc, 1000-1), and toluene (9200-1) were purchased from Caledon Laboratory Chemicals. Sodium hypochlorite (NaOCl) solution (239305), silver chloride (AgCl, 227927), poly(vinyl chloride) (PVC, 389293), cyclohexanone (C102180), phosphoric acid (H_3PO_4 , 695017), boric acid (H_3BO_3 , 339067), phosphate buffered saline (PBS) tablet (P4417), sodium chloride (NaCl, 793566), sodium acetate (CH_3COONa , S2889), sodium phosphate dibasic (Na_2HPO_4 , S0876), potassium

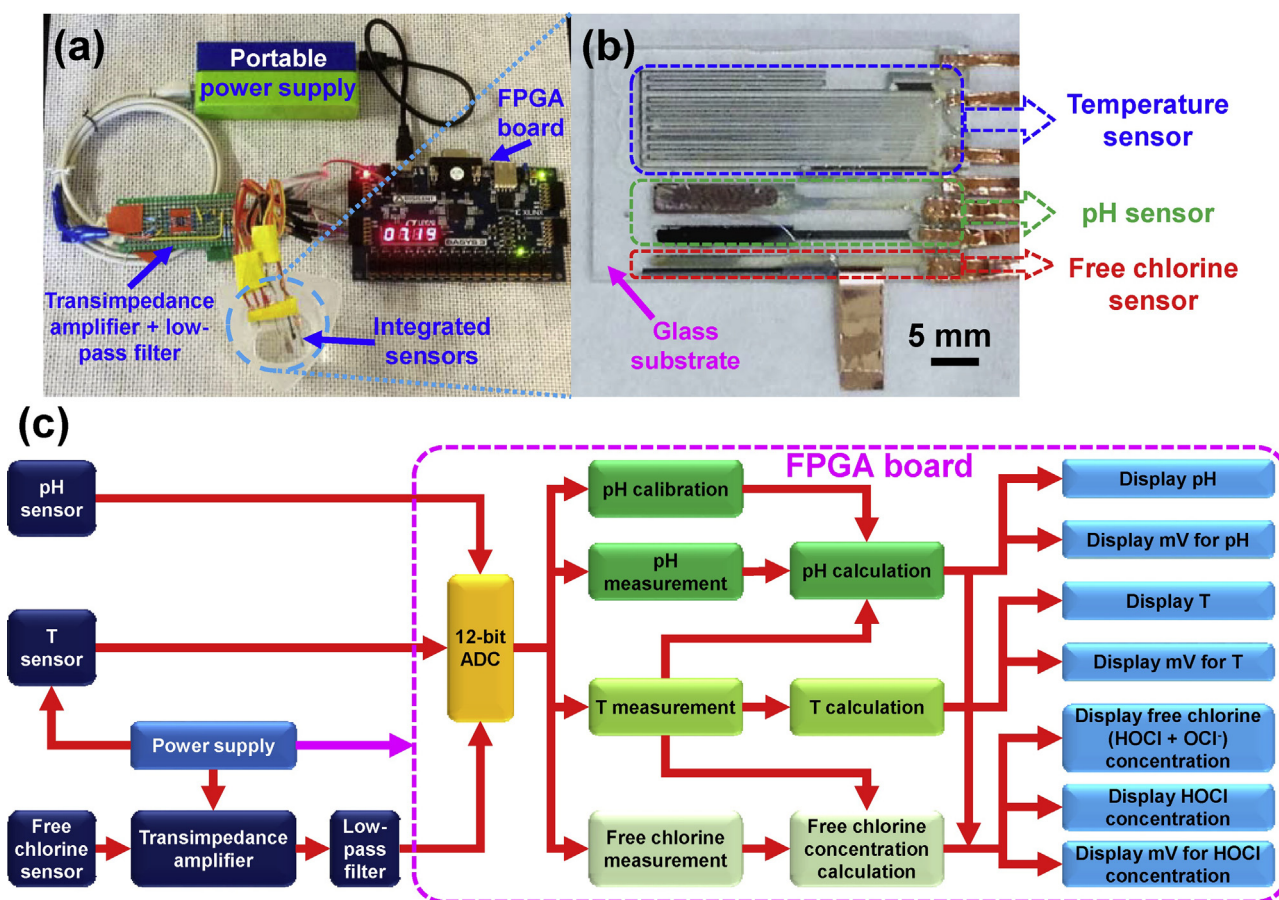


Fig. 2. (a) A portable integrated water quality monitoring system measuring the pH value of tap water. (b) An integrated water quality sensing probe with pH, temperature and free chlorine sensors. (c) A schematic diagram of the water quality monitoring system including signal conditioning (amplification and filtering for free chlorine sensor), data analysis (on the FPGA board), and results display (on the FPGA board). “T” represents “temperature” in the schematic diagram.

sulfate (K_2SO_4 , 221325), ammonium carbamate (292834), propylene glycol monomethyl ether acetate (PGMEA, 484431), and sodium hydroxide (NaOH) pellets (S5881) were from Sigma-Aldrich. Pd precursor solution (Pd-25c) and silver nanoparticle (AgNP) ink (xcm-ns1j) were obtained from Xerox Research Centre of Canada. Potassium chloride (KCl, PX1405), sodium carbonate (Na_2CO_3 , SX0395-1), sodium bicarbonate (NaHCO_3 , SX0320-1), ammonium sulfate ($(\text{NH}_4)_2\text{SO}_4$, AX1385-3), and magnesium chloride (MgCl_2 , MX0045-2) were from EMD Millipore. Potassium nitrate (KNO_3 , 74336-300) was from Anachemia. Calcium chloride (CaCl_2 , C77-500) was from Fisher Scientific. SU-8 3035 photoresist was purchased from MicroChem.

2.2. Fabrication and integration processes

The integrated pH, free chlorine, and temperature sensors were fabricated on glass substrates (7525 M, J. Melvin Freed Brand) cleaned with IPA and de-ionized (DI) water (Fig. 1(a)). The pH and temperature sensors were inkjet-printed onto the glass substrates while the free chlorine sensors were fabricated separately by the electrochemical modification of pencil leads.

The Pd ink was prepared by diluting the as-received Pd precursor solution using 20 wt% toluene. The inkjet printing process (Fig. 1(b)) followed our previous report [21] using a Dimatix DMP-2831 materials printer (Fujifilm) equipped with DMC-11610 Dimatix materials cartridges (Fujifilm). The thermolysis of the printed Pd ink was carried out on a hotplate in air at 200°C for 4 min to form a smooth Pd film (Fig. 1(c)). Then, the Pd ink was printed again, and the thermolysis was performed in a low-vacuum cham-

ber (Isotemp 280A, Fisher Scientific) at 200°C for 20 min to form rough Pd films (Fig. 1(d–e)) [24]. The printed Pd films (both smooth and rough) were oxidized in air at 200°C for 48 h to form PdO-rich films (Fig. 1(f)). The smooth film is used as the sensing electrode in the pH sensor, and the rough films are used as two thermistors with a negative temperature coefficient of resistance (TCR) in the Wheatstone-bridge-based temperature sensor.

Next, the SU-8 photoresist was diluted by 10 times using PGMEA and inkjet-printed onto the substrate (Fig. 1(g)) to improve the adhesion between the substrate and the Ag films (printed in the next step). The SU-8 ink was cured in air at 200°C for 1 h (Fig. 1(h)). The as-received AgNP ink was then inkjet-printed to form two thermistors with positive TCR values in the temperature sensor, the Ag layer of the reference electrode in the pH sensor, and the Ag electrode in the free chlorine sensor (Fig. 1(i)). The sintering condition for the AgNP ink was 120°C for 10 min in air (Fig. 1(j)). The AgCl layer in the Ag/AgCl reference electrode was obtained by drop casting 0.4 wt% NaOCl solution onto the printed Ag for 30 s before rinsing with DI water (Fig. 1(k)). A solution of the solid electrolyte of the reference electrode was formulated by dissolving 2 wt% PVC in KCl- and AgCl-saturated cyclohexanone. The solution was drop casted on top of the Ag/AgCl film and heated at 140°C for 5 min to form a solid electrolyte layer (Fig. 1(l)).

The sensing electrode of the free chlorine sensor was fabricated by the electrochemical modification of the pencil lead surface [34]. A voltage of 1 V was applied between the pencil lead and a Ag/AgCl reference electrode (CHI111, CH Instruments). The electrolyte solution contained 0.1 M PBS (pH 7.0) and an appropriate amount of ammonium carbamate (0.1 M) to adjust the pH to 8.9. The middle

part of the modified pencil lead was mounted onto the glass substrate by using a 2 wt% PVC solution in cyclohexanone as adhesive (Fig. 1(m–n)). The PVC solution also covered the entire temperature sensor and the printed electrical connections of the pH and free chlorine sensors as a water-resistant layer (Fig. 1(n)). After drying the PVC solution in air at 80 °C for 30 min, conductive copper tapes (Tapes Master) were used to make electrical contacts. The integrated sensor was conditioned in 1 M KCl solution for 12 h before use.

The fabricated sensors were connected to a FPGA board (Basys 3, Artix-7, Digilent) for data analysis and display. The FPGA was programmed using the Verilog language with the Vivado Design Suite (2015.4, Xilinx).

2.3. Sensor characterization

2.3.1. pH sensing

Britton–Robinson pH buffer solutions (pH = 4, 5, 6, 7, 8, 9, 10) were prepared by mixing a solution comprising 0.04 M H_3PO_4 , 0.04 M HOAc , 0.04 M H_3BO_3 with 0.2 M NaOH solution. The pH values of the buffer solutions were monitored by a commercial pH meter (PHB-600R, OMEGA) with a glass electrode (PHE1311, OMEGA). The performance of the inkjet-printed pH sensors was characterized based on a potentiometric configuration using a semiconductor analyzer (4200-SCS, Keithley). The open-circuit potential between the electrodes was recorded every 2 s for 90 s when the electrodes were immersed in pH buffer solutions (without stirring) [35]. The electrodes were transferred into the next pH buffer solution without cleaning or drying. All pH sensing tests were performed at 27 ± 2 °C.

2.3.2. Free chlorine sensing

Free chlorine solutions with different concentrations were prepared by diluting the as-received NaOCl solution using a PBS solution (0.01 M, pH = 7.4). The free chlorine concentrations of the prepared solutions were calibrated using a DPD-based colorimetric test kit (CN-70, Hach). The performance of the free chlorine sensors was characterized based on a simplified amperometric configuration: the modified pencil leads as the working electrodes and a printed Ag film as the reference/counter electrode. The current flowing through the working and the reference/counter electrodes was recorded using a semiconductor parameter analyzer (4200-SCS, Keithley) every 2 s for 60 s when the electrodes were immersed in free chlorine solutions (without stirring). The electrodes were transferred into the next free chlorine solution without cleaning or drying. All free chlorine sensing tests were performed at 27 ± 2 °C.

2.3.3. Temperature sensing

The performance the Wheatstone-bridge-based temperature sensors were characterized in a reliability test chamber (ESL-2CA, ESPEC) in water with the temperature varying between 0 °C and 50 °C. The water temperature was monitored by a commercial thermometer (HI98509 Checktemp 1, HANNA). The bias voltage for the temperature sensors was 3.3 V and the output voltage was recorded using a semiconductor parameter analyzer (4200-SCS, Keithley).

3. Results and discussion

3.1. System design

The developed water quality monitoring system consists of an integrated sensor probe, a signal conditioning circuit board for free chlorine sensing, a FPGA board, and a portable power supply (Fig. 2(a)). The integrated sensor probe includes a pH sensor, a free chlorine sensor, and a temperature sensor (Fig. 2(b)). Here, the sensor signal was digitized using a 12-bit analogue-to-digital converter

(ADC, on the FPGA board), whose input voltage range was 0–1 V (reference voltage = 1.25 V) with a resolution of 0.305 mV. The data sampling frequency was set to be 1 Hz to realize real-time monitoring. The digitized signal was sent to the microprocessor (on the FPGA board) programmed with the following functions for data analyses and results display (Fig. 2(c)):

- (1) Temperature measurement. The output voltage of the temperature sensor is recorded in real-time. The temperature is calculated using a programmed equation (see details in Section 3.4).
- (2) pH calibration. The stabilized output voltage of the pH sensor is stored as the calibration voltage when the sensor is immersed in a standard calibration solution of pH = 7.
- (3) pH measurement. The output voltage of the pH sensor is recorded in real-time. The pH value is calculated using the sensitivity of the sensor and the calibration voltage. The pH sensitivity is compensated using the measured temperature (see details in Section 3.2).
- (4) Free chlorine measurement. The output voltage of the free chlorine sensor is recorded in real-time. The concentration of HOCl is calculated using an equation programmed in the FPGA. The HOCl sensitivity is compensated using the measured temperature. Also, to calculate the free chlorine concentration (both HOCl and OCl^-), another programmed equation is used based on the measured pH and temperature of the water sample (see details in Section 3.3).

Monitoring the water quality using the developed integrated sensing system has two sets of important advantages over using discrete sensors and conventional analytical methods:

- (1) The integrated system is easy-to-use and outputs three important water quality parameters: pH, free chlorine concentration, and temperature. The user does not have to take readings from three different sensors and manually perform the signal compensation. The integrated system allows simultaneous on-site monitoring with real-time compensation for a high measurement accuracy. For example, the pH reading is compensated using the measured temperature. The calculated free chlorine concentration is compensated using both pH and temperature.
- (2) The materials used for fabricating the pH and temperature sensors are the same. These sensing materials are deposited using the same/similar methods to reduce the fabrication complexity. The major technology used to fabricate the sensors is inkjet printing, so the integrated sensor is low-cost due to the small amount of consumed materials and the elimination of cleanroom environment, high-temperature equipment, high-vacuum chambers, and physical masks for patterning.

The FPGA board is used as the electronics part in this study for proof of concept and fast prototyping. In future, the integration level of the entire system, especially the electronics part, can be enhanced for smaller system dimensions and more convenient practical usage. For example, the transimpedance amplifier, low-pass filter, ADC, microprocessor, and memory units can be integrated on a single silicon chip using microelectronics technologies. The cost of such a highly integrated system can be further reduced compared to the present system. Moreover, the jumper connections between the sensors and the electronics will be replaced by plug-in type connectors in the future system to reduce the signal interference.

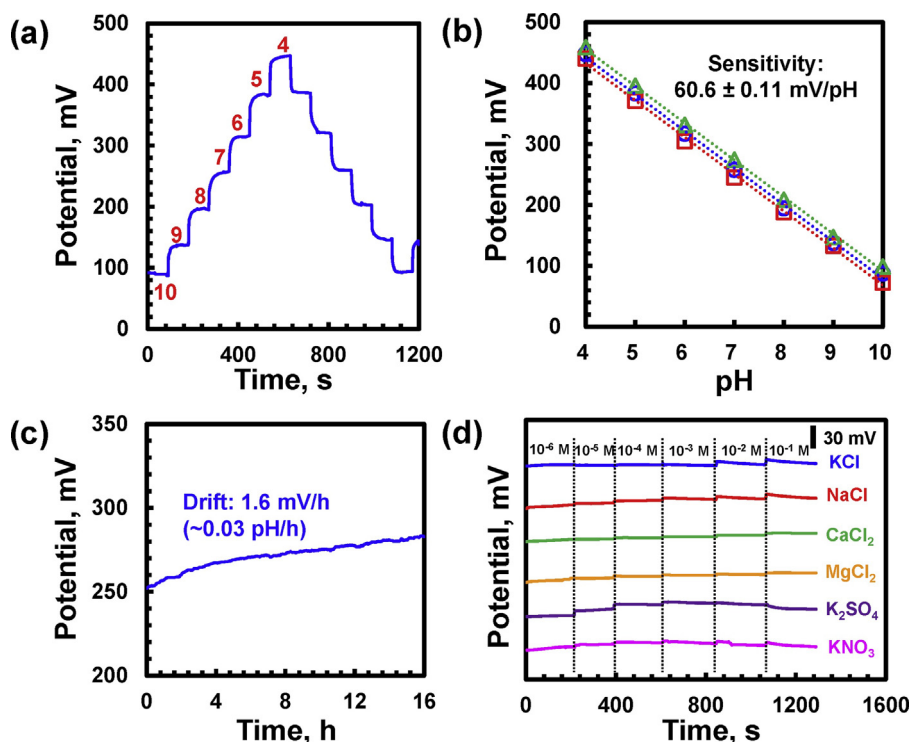
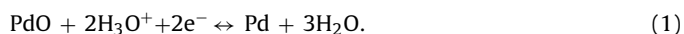


Fig. 3. (a) Temporal response of an inkjet-printed pH sensor for pH between 4 and 10. (b) Calibration curves for three printed pH sensors. (c) Drift behavior of a printed pH sensor in 1 M KCl solution in dark. (d) The response of a printed pH sensor in interfering solutions with different concentrations.

3.2. Characterization of pH sensor

The potentiometric pH sensor used in the integrated system consisted of an inkjet-printed Pd/PdO sensing electrode and an inkjet-printed Ag/AgCl/KCl reference electrode [21]. Pd/PdO was chosen as the pH sensing material due to its higher sensitivity and stability than other materials such as copper/copper oxide, ruthenium/ruthenium oxide, polyaniline, and carbon-based materials [5]. Also, the Pd/PdO-based pH sensing electrode is low cost (<1 cent), because <1 μ L Pd ink is consumed in the fabrication process. The pH sensing mechanism of the Pd/PdO electrode can be described by the following redox reaction [35]:



According to the Nernst equation, the redox potential can be calculated by:

$$E = E^0 - \frac{2.303RT}{F} \text{pH}, \quad (2)$$

where E^0 is the standard electrode potential, R is the gas constant (8.31 J/mol/K), T is the absolute temperature, and F is Faraday's constant (96485.33C/mol). When the pH of the water sample varied between 4 and 10, the sensor exhibited a large, stable, and low-noise output voltage in the range of 0–500 mV (within the input range of the ADC). The output of such a sensor was within the ADC voltage range in the electronic system so that a power supply, amplifier, or low-pass filter was not required for driving the sensor or signal conditioning. Also, the sensor showed a short response time of ~ 15 s (Fig. 3(a)). This fast response allowed the implementation of the sensor for almost real-time monitoring.

In Fig. 3(b), the sensitivity values of three pH sensors were ~ 60.6 mV/pH, which was close to the theoretical Nernstian response (59.53 mV/pH at 27 °C). Also, the sensor-to-sensor (3 sensors) variation of the sensitivity was as small as ± 0.11 mV/pH ($\sim 0.36\%$ of the sensitivity). However, the standard electrode potential (E^0 , or the output voltage for a certain pH value) of different

sensors exhibited a difference of ~ 30 mV (Fig. 3(b)), which corresponds to a difference of ~ 0.5 pH unit. Thus, single-point calibration (normally at pH = 7) should be carried out for each sensor to find its standard electrode potential (E^0) for improved sensing accuracy. Moreover, in Eq. (2), the pH sensitivity is a function of temperature. To experimentally verify this dependence, the sensitivity of the pH sensor was measured at 3 °C, 15 °C, 27 °C and 40 °C (Fig. S2(a), Supporting Information). The pH sensitivity showed a linear temperature dependence of ~ 0.23 mV/pH/°C. Thus, temperature compensation for the pH sensitivity is required to improve the accuracy of a pH measurement. The equation for calculating the pH value can be written as:

$$\text{pH} = 7 + \frac{E_{\text{cal}} - E_{\text{meas}}}{60.6 + (T_{\text{meas}} - 27) \times 0.23}, \quad (3)$$

and this equation was programmed into the microprocessor on the FPGA board. In Eq. (3), E_{cal} (in mV) is the recorded voltage when the sensor is immersed into the calibration solution (pH = 7), E_{meas} (in mV) is voltage reading during pH monitoring, T_{meas} (in °C) is the measured temperature of the water sample.

The resolution of the sensor is determined by its hysteresis. When the pH was cycled between 4 and 10 (Fig. 3(a)), the average hysteresis of 8.9 mV resulted in a resolution of 0.15 pH for the sensor. The drift behavior of the sensor was also studied for evaluating the long-term stability. The printed sensor was kept in 1 M KCl solution (to reduce the drift of the Ag/AgCl/KCl reference electrode) at room temperature in dark for 16 h. The drift rate was ~ 1.6 mV/h (Fig. 3(c)), which was equivalent to 0.03 pH/h. The drift-induced inaccuracy was 5 times lower than the resolution of the sensor, so time compensation is not needed for most on-demand water quality monitoring applications using this fast-response sensor. In addition, the selectivity was studied by immersing the sensor into solutions containing different concentrations of interfering ions (e.g. K^+ , Na^+ , Ca^{2+} , Mg^{2+} , Cl^- , SO_4^{2-} , NO_3^-). The sensor had negligible response to common anions and cations (Fig. 3(d)) due to the specific redox reaction shown in Eq. (1). We believe the only

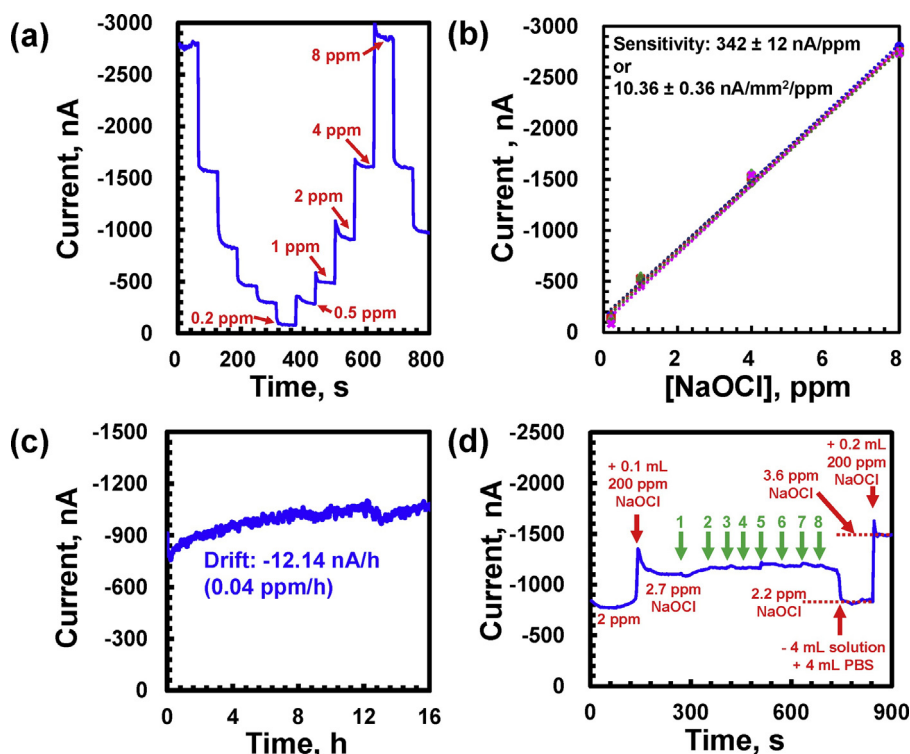


Fig. 4. (a) Temporal response of a free chlorine sensor for free chlorine concentrations between 0.2 and 8 ppm. (b) Calibration curves for five free chlorine sensors. (c) Drift behavior of a free chlorine sensor in 2 ppm NaOCl solution in dark. (d) The response of a free chlorine sensor to NaOCl solutions and 400 ppm interfering solutions. Each green arrow indicates the addition of 0.2 mL of 5% interfering solution to 25 mL NaOCl solution (1, KNO_3 ; 2, K_2SO_4 ; 3, Na_2CO_3 ; 4, NaHCO_3 ; 5, NaCl ; 6, $(\text{NH}_4)_2\text{SO}_4$; 7, NaH_2PO_4 ; 8, NaOAc). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

minor voltage variations shown in Fig. 3(d) were attributed to the exchange of K^+ or Cl^- between the solutions and the solid electrolyte layer on the printed reference electrode.

In summary, the developed pH sensor has the following advantages compared to conventional glass electrodes and other pH sensors/sensing systems fabricated by microelectronics technologies [5,36]:

- (1) The sensor has a higher sensitivity (~ 60 mV/pH) than many ion-selective field-effect transistors (~ 55 mV/pH).
- (2) The sensor is easier to handle and use, because a reference electrode is integrated onto the same substrate, and the sensor can be calibrated at a single pH value due to its reproducible sensitivity and linear response.
- (3) The sensor fabricated by inkjet printing technology is low-cost.
- (4) An external power supply, amplifier, or filter is not needed, which simplifies the integration and signal processing.

3.3. Characterization of free chlorine sensor

The free chlorine concentration was monitored by a simplified amperometric sensor with an amine-modified pencil lead as the working electrode [34]. At the working electrode, HOCl undergoes electrochemical reduction according to:



and the output current is proportional to the HOCl concentration. Because of the high complexity of integrating a conventional amperometric sensor (3 electrodes and a potentiostat), the reference and counter electrodes in our sensor were combined as an inkjet-printed Ag film. This 2-electrode system also did not require an external power supply (potentiostat) to activate the sensor [28], since 0 V was used as the reduction potential. This sensor mea-

sures the concentration of HOCl in the water sample [34], and the result can be used to calculate the free chlorine concentration (both HOCl and OCl^-). When the free chlorine concentration of the water sample varied between 0.2 and 8 ppm, the amplitude of the output current of the sensor was between 0 and 3000 nA (Fig. 4(a)). A transimpedance amplifier with a gain of 196 k Ω was designed to convert the current signal to a voltage signal between 0 and 1 V (Fig. S1, Supporting Information). To suppress the noise introduced by the transimpedance amplifier, a resistor-capacitor low-pass filter with a cutoff frequency of 0.06 Hz was used for signal conditioning.

In Fig. 4(a), we find the response time of the free chlorine sensor was ~ 30 s, which is suitable for on-site monitoring. Next, the sensitivities of 5 free chlorine sensors (fabricated in the same batch) were measured (Fig. 4(b)). The average sensitivity was ~ 342 nA/ppm (10.36 nA/mm²/ppm for the sensors with an electrode area of ~ 33 mm²) and the sensitivity variation was only ± 12 nA/ppm (0.36 nA/mm²/ppm, $\sim 7\%$ of the sensitivity). Although the sensitivity was not as high as that of other reported free chlorine sensors [5], our sensor had a high reproducibility with a negligible sensor-to-sensor variation of the output current at a certain free chlorine concentration. This high reproducibility was appealing for an easy-to-use sensor because calibration is not needed before each measurement once the calibration equation was stored in the electronic system. To improve the sensitivity of the free chlorine sensor, nano-structured carbon materials with a large surface area (e.g. carbon nanotubes, graphene, and nanoporous carbon) can be used as the working electrode [37]. Thus, a larger amount of amine-based reactive sites can be immobilized and react with HOCl for a more sensitive output.

To study the temperature dependence of the sensitivity, the sensitivity of the free chlorine sensor was measured at 3 °C, 15 °C, 27 °C and 40 °C (Fig. S2(c), Supporting Information). The free chlorine sensitivity showed a linear temperature dependence

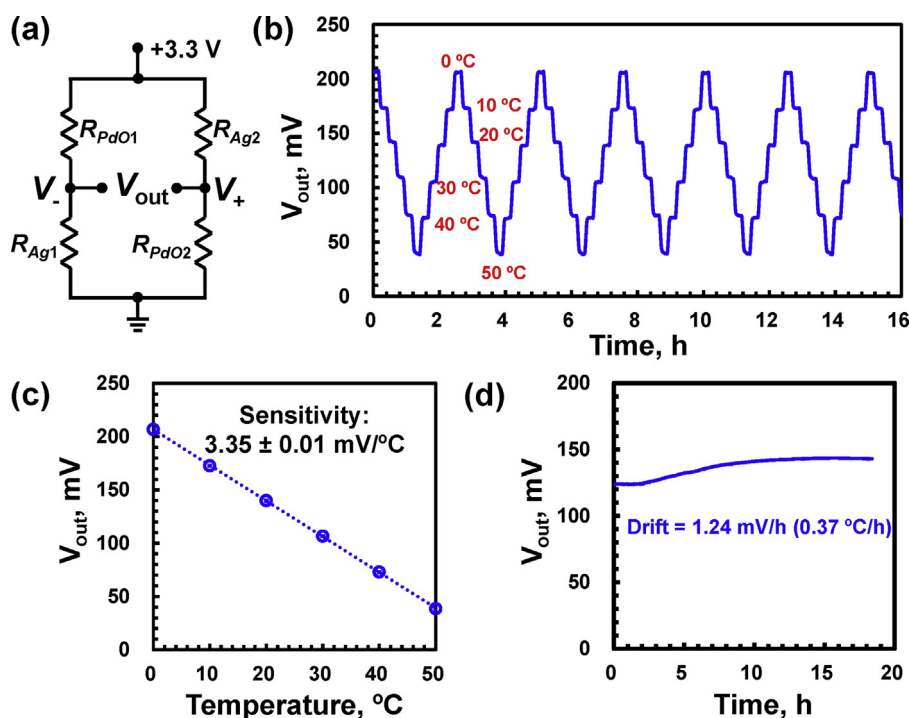


Fig. 5. (a) Circuit diagram of an inkjet-printed Wheatstone-bridge-based temperature sensor with 4 thermistors. (b) Temporal response of the temperature sensor for temperatures between 0 and 50 °C. (c) Calibration curve of the temperature sensor. (d) Drift behavior of the temperature sensor at 25 °C in water in dark.

of $\sim 9.3 \text{ nA/ppm/}^\circ\text{C}$. Thus, temperature compensation for the free chlorine sensitivity is required to improve the accuracy of a free chlorine measurement. The calibration equation of the free chlorine sensor can be written as:

$$I_{out} = [342 + (T_{meas} - 27) \times 9.3] C_{NaOCl} + 109.6, \quad (5)$$

where I_{out} (in nA) is the amplitude of the output current of the free chlorine sensor (input current of the signal conditioning circuit), T_{meas} (in °C) is the measured temperature of the water sample, and C_{NaOCl} (in ppm) is the concentration of NaOCl in the water sample for sensor characterization. The transfer function of the signal conditioning circuit (transimpedance amplifier and low-pass filter) can be described by:

$$V = 0.196 I_{out} + 35.4, \quad (6)$$

where V (in mV) is the output voltage of the signal conditioning circuit. Thus, considering the pH and temperature of the water sample, Eq. (7) (see Supporting Information for detailed derivation) was programmed into the microprocessor to calculate the free chlorine concentration:

$$C_{free \text{ chlorine}} = 0.57 \times \frac{\frac{V-35.4}{0.196} - 109.6}{342 + (T_{meas} - 27) \times 9.3} \times \left\{ 1 + 10^{pH - \left[\frac{3000}{T_{meas} - 273} - 10.0686 + 0.0253(T_{meas} + 273) \right]} \right\}. \quad (7)$$

The resolution of the free chlorine sensor is determined by its hysteresis. When the free chlorine concentration was cycled from 0.2 to 8, and back to 0.2 ppm (Fig. 4(a)), the average hysteresis of 53.5 nA resulted in a resolution of 0.16 ppm. The drift behavior was studied by keeping the free chlorine sensor in 2 ppm NaOCl solution at 4 °C in dark (to reduce the decomposition of free chlorine) for 16 h. The drift rate was $\sim 12.14 \text{ nA/h}$ (Fig. 4(c)), which was equivalent to 0.04 ppm/h (4 times lower than the resolution of the sensor). In addition, the selectivity was studied by recording the sensor response when 0.2 mL of 5% interfering solutions were added into

25 mL of 2 ppm NaOCl solution (Fig. 4(d)). The added interfering solutions had an equivalent concentration of $\sim 400 \text{ ppm}$. The sensor had negligible response to KNO_3 , K_2SO_4 , Na_2CO_3 , NaHCO_3 , NaCl , $(\text{NH}_4)_2\text{SO}_4$, NaHPO_4 and NaOAc . After the selectivity test, the sensor was still able to show a stable response to the change of free chlorine concentration.

In summary, the developed free chlorine sensor has the following advantages compared to conventional amperometric sensors and other free chlorine sensors fabricated by microelectronics technologies [5,38]:

- (1) Although the sensitivity of the free chlorine sensor is relatively low, its sensing accuracy can be high since it can be integrated with pH and temperature sensors for signal compensation.
- (2) The sensor is highly reproducible, and calibration is only required for one sensor before the first use (calibration-free for other sensors fabricated in the same batch).
- (3) A potentiostat is not needed for activating the sensor, which simplifies the integration.
- (4) Sample preconditioning steps (e.g. oxidation, dilution, pH adjustment) using chemicals are not needed, which reduces the operation complexity of the user.
- (5) The sensor is cost-effective due to the use of pencil lead and inkjet-printed Ag film.

3.4. Characterization of temperature sensor

For the temperature sensor, we used a Wheatstone bridge circuit (Fig. 5(a)). In this circuit, $R_{Ag1,2}$ were inkjet-printed Ag films that have a positive TCR value of $0.197\%/^\circ\text{C}$ (Fig. S3(a–b), Supporting Information). $R_{PdO1,2}$ were inkjet-printed Pd/PdO films with a bilayer morphology that have a negative TCR value of $-0.256\%/^\circ\text{C}$ (Fig. S3(c–d), Supporting Information) [24]. The selected thermistor materials of Ag and PdO were the same for the fabrication of the pH sensor. Hence, the temperature sensor could be simultaneously fabricated with the pH sensor, thus simplifying the integration

Table 1
Monitoring of pH, free chlorine concentration and temperature of real water samples (average results of 5 measurements).

	Tap water (Toronto)		Lake water (Lake Ontario)		Swimming pool water (McMaster University)	
	Our system	Reference	Our system	Reference	Our system	Reference
pH	8.30 ± 0.29	8.11 ± 0.07	7.49 ± 0.10	7.88 ± 0.03	6.88 ± 0.10	7.36 ± 0.10
Free chlorine, ppm	0.95 ± 0.13	0.86 ± 0.13	0.00 ± 0.00	0.00 ± 0.00	2.62 ± 0.32	2.22 ± 0.22
Temperature, °C	14.70 ± 0.52	15.04 ± 0.11	0.20 ± 0.08	0.24 ± 0.05	26.34 ± 0.53	27.72 ± 0.08

process. The resistance values of the four thermistors in the Wheatstone bridge circuit were chosen to be $R_{Ag1} = 150 \Omega$, $R_{Ag2} = 525 \Omega$, $R_{PdO1} = 675 \Omega$, and $R_{PdO2} = 150 \Omega$ (Fig. 5(a)) to satisfy the following conditions:

- (1) The output voltage is in the range of 0–1 V to be read by the ADC for a temperature between 0 °C and 50 °C (for most water quality monitoring applications).
- (2) The sensitivity of the temperature sensor is maximized for high-accuracy measurements.
- (3) The resistance between the power supply and ground should be large so that the self-heating of the resistors is kept low to reduce drift of the sensor output.
- (4) The size of the sensor is kept small to obtain a high spatial resolution.

The temporal response of the temperature sensor was measured between 0 and 50 °C in water. When the temperature increases, the resistance of R_{Ag1} increases and the resistance of R_{PdO1} decreases. Thus, the voltage of V_- increases. Similarly, the voltage of V_+ decreases ($V_{out} = V_+ - V_-$ also decreases) as the temperature increases. Using this differential sensing concept, the temperature sensitivity, dV_{out}/dT , of the Wheatstone bridge is increased compared with a thermistor. In addition, the output voltage showed a fast (almost immediate) and linear response with the temperature change (Fig. 5(b)), so the sensor is capable for real-time compensation of the pH and free chlorine sensors.

From the temporal response, the sensitivity of the temperature sensor was calculated to be $\sim 3.35 \text{ mV}/^\circ\text{C}$ (Fig. 5(c)), which was lower than the theoretical value ($\sim 4.3 \text{ mV}/^\circ\text{C}$). This reduced sensitivity might be caused by the PVC coating on the 4 thermistors. Because PVC (52 ppm/°C) has a different thermal expansion coefficient than Ag (19 ppm/°C) and PdO (12 ppm/°C), the temperature variation induced stress could cause resistance increase of the 4 thermistors, resulting in a reduced sensitivity. The equation to be programmed into the microprocessor for temperature calculation can be written as:

$$T_{meas} = \frac{206.7 - V_{out}}{3.35}, \quad (8)$$

where T_{meas} (in °C) is the measured temperature of the water sample and V_{out} (in mV) is the output voltage of the temperature sensor.

The temperature sensor exhibited a hysteresis of $\sim 0.93 \text{ mV}$ (Fig. 5(b)), so the resolution of the sensor was $\sim 0.28^\circ\text{C}$. The drift behavior of the sensor was studied by storing the sensor at 25 °C in water in dark. Over the total test time of $\sim 18 \text{ h}$, the drift rate was 1.24 mV/h (0.37°C/h). In Fig. 5(d), the drift behavior can be divided into two segments. The drift rate of $\sim 2.07 \text{ mV/h}$ ($\sim 0.62^\circ\text{C/h}$) was relatively large for the first 10 h, which might be due to the water absorption of the PVC film. The swelling of the PVC film resulted in stress on the underlying thermistors so that their resistance drifted. After 10 h of the test, the drift rate decreased to $\sim 0.23 \text{ mV/h}$ ($\sim 0.09^\circ\text{C/h}$). This slow drift suggested that an equilibrium condition was reached for the swelling of PVC in water. To reduce the drift, the temperature sensor can be encapsulated by a polymer

with a lower water absorption rate such as parylene-C or liquid crystal polymer [39].

In summary, the developed temperature sensor has the following advantages:

- (1) The Wheatstone-bridge-based temperature sensor has a higher sensitivity than a thermistor.
- (2) The sensor is fabricated by inkjet printing using the same materials as fabricating the pH sensor, so the sensor is inexpensive and easy-to-integrate.

3.5. Monitoring of real water samples

The integrated sensors were connected to the programmed FPGA board for the monitoring of tap water, lake water, and swimming pool water. The pH, free chlorine, temperature readings from our developed system were compared with results from reference methods (Table 1). For pH sensing, a commercial pH meter with a glass electrode was used as the reference. For free chlorine sensing, a DPD-based colorimetric test kit was used as the reference. For temperature sensing, a thermometer was used as the reference. The results obtained using our system were close to those measured from reference methods, with a difference less than 7%, 18%, and 15% for pH, free chlorine, and temperature, respectively.

4. Conclusions

We have developed an integrated sensing system for the simultaneous monitoring of water quality parameters including pH, free chlorine concentration, and temperature. The system consisted of an integrated sensing probe with inkjet-printed pH and temperature sensors, and an electrochemically modified free chlorine sensor. These sensors were compact in size (total size $\sim 25 \text{ mm} \times 30 \text{ mm}$), highly sensitive, fast-in-response, user-friendly, and cost-effective. A measurement accuracy of $>82\%$ was obtained for the monitoring of real water samples using the sensing probe with a programmed FPGA board. This study signifies the practical application of the integrated sensing system as a simplification of the conventional laboratory-based analytical methods for on-site water quality monitoring. The developed platform can also be exploited for the monitoring of other water quality parameters such as conductivity, heavy metal ions, dissolved oxygen, and bacteria.

Acknowledgements

This research is supported by Discovery Grants from the Natural Science and Engineering Research Council of Canada, an infrastructure grant from the Canada Foundation for Innovation, a FedDev of Southern Ontario grant, the Canada Research Chair program, NSERC ResEau strategic network and the NCE IC-IMPACTS.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.snb.2017.07.188>.

References

- [1] World Health Organization, Guidelines for Drinking-water Quality, 4th edition, 2011.
- [2] D. Chapman, Water Quality Assessments: A Guide to the Use of Biota, Sediments, and Water in Environmental Monitoring, University Press Cambridge, UK, 1996.
- [3] M.V. Storey, B. van der Gaag, B.P. Burns, Advances in on-line drinking water quality monitoring and early warning systems, *Water Res.* 45 (2011) 741–747, <http://dx.doi.org/10.1016/j.watres.2010.08.049>.
- [4] M.H. Banna, S. Imran, A. Francisque, H. Najjaran, R. Sadiq, M. Rodriguez, et al., Online drinking water quality monitoring: review on available and emerging technologies, *Crit. Rev. Environ. Sci. Technol.* 44 (2014) 1370–1421, <http://dx.doi.org/10.1080/10643389.2013.781936>.
- [5] Y. Qin, H.-J. Kwon, M.M.R. Howlader, M.J. Deen, Microfabricated electrochemical pH and free chlorine sensors for water quality monitoring: recent advances and research challenges, *RSC Adv.* 5 (2015) 69086–69109, <http://dx.doi.org/10.1039/C5RA11291E>.
- [6] USEPA, Water Sentinel Online Water Quality Monitoring as an Indicator of Drinking Water Contamination, EPA 817-D-05-002, U.S. Environmental Protection Agency, Water Security Division, Washington, D.C., 2005.
- [7] Y. Feng, D.W. Smith, J.R. Bolton, Photolysis of aqueous free chlorine species (HOCl and OCl⁻) with 254 nm ultraviolet light, *J. Environ. Eng. Sci.* 6 (2007) 277–284, <http://dx.doi.org/10.1139/s06-052>.
- [8] S. Joo, R.B. Brown, Chemical sensors with integrated electronics, *Chem. Rev.* 108 (2008) 638–651, <http://dx.doi.org/10.1021/cr068113+>.
- [9] M. Badihi-Mossberg, V. Buchner, J. Rishpon, Electrochemical biosensors for pollutants in the environment, *Electroanalysis* 19 (2007) 2015–2028, <http://dx.doi.org/10.1002/elan.200703946>.
- [10] Y. Xia, J. Si, Z. Li, Fabrication techniques for microfluidic paper-based analytical devices and their applications for biological testing: a review, *Biosens. Bioelectron.* 77 (2016) 774–789, <http://dx.doi.org/10.1016/j.bios.2015.10.032>.
- [11] G. Connell, Water Disinfection Series: The Chlorination/Chloramination Handbook, American Water Works Association, Denver, CO USA, 1996.
- [12] K. Senthilkumar, J.-M. Zen, Free chlorine detection based on EC⁺ mechanism at an electroactive polymelamine-modified electrode, *Electrochem. Commun.* 46 (2014) 87–90, <http://dx.doi.org/10.1016/j.elecom.2014.06.018>.
- [13] Y. Qin, S. Pan, M.M.R. Howlader, R. Ghosh, N.-X. Hu, M.J. Deen, Paper-based, hand-drawn free chlorine sensor with poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate), *Anal. Chem.* 88 (2016) 10384–10389, <http://dx.doi.org/10.1021/acs.analchem.6b03211>.
- [14] J.C. Morris, The acid ionization constant of HOCl from 5 to 35°, *J. Phys. Chem.* 70 (1966) 3798–3805, <http://dx.doi.org/10.1021/j100884a007>.
- [15] G. Gordon, D.L. Sweetin, K. Smith, G.E. Pacey, Improvements in the N,N-diethyl-p-phenylenediamine method for the determination of free and combined residual chlorine through the use of FIA, *Talanta* 38 (1991) 145–149, [http://dx.doi.org/10.1016/0039-9140\(91\)80122-G](http://dx.doi.org/10.1016/0039-9140(91)80122-G).
- [16] P. Salazar, M. Martín, F.J. García-García, J.L. González-Mora, A.R. González-Elipe, A novel and improved surfactant-modified Prussian Blue electrode for amperometric detection of free chlorine in water, *Sens. Actuator B Chem.* 213 (2015) 116–123, <http://dx.doi.org/10.1016/j.snb.2015.02.092>.
- [17] P. Salazar, M. Martín, J.L. González-Mora, A.R. González-Elipe, Application of Prussian Blue electrodes for amperometric detection of free chlorine in water samples using flow injection analysis, *Talanta* 146 (2016) 410–416, <http://dx.doi.org/10.1016/j.talanta.2015.08.072>.
- [18] L.H.H. Hsu, E. Hoque, P. Kruse, P. Ravi Selvaganapathy, A carbon nanotube based resettable sensor for measuring free chlorine in drinking water, *Appl. Phys. Lett.* 106 (2015) 63102, <http://dx.doi.org/10.1063/1.4907631>.
- [19] A. Richter, G. Paschew, S. Klatt, J. Lienig, K.-F. Arndt, H.-J.P. Adler, Review on hydrogel-based pH sensors and microsystems, *Sensors* 8 (2008) 561–581, <http://dx.doi.org/10.3390/s8010561>.
- [20] D. Wencel, T. Abel, C. McDonagh, Optical chemical pH sensors, *Anal. Chem.* 86 (2014) 15–29, <http://dx.doi.org/10.1021/ac4035168>.
- [21] Y. Qin, A.U. Alam, M.M.R. Howlader, N.-X. Hu, M.J. Deen, Inkjet printing of a highly loaded palladium ink for integrated, low-cost pH sensors, *Adv. Funct. Mater.* 26 (2016) 4923–4933, <http://dx.doi.org/10.1002/adfm.201600657>.
- [22] D.-H. Kim, N. Lu, R. Ghaffari, Y.-S. Kim, S.P. Lee, L. Xu, et al., Materials for multifunctional balloon catheters with capabilities in cardiac electrophysiological mapping and ablation therapy, *Nat. Mater.* 10 (2011) 316–323, <http://dx.doi.org/10.1038/nmat2971>.
- [23] M. Sezen, J.T. Register, Y. Yao, B. Glisic, Y.-L. Loo, Eliminating piezoresistivity in flexible conducting polymers for accurate temperature sensing under dynamic mechanical deformations, *Small* 12 (2016) 2832–2838, <http://dx.doi.org/10.1002/smll.201600858>.
- [24] Y. Qin, A.U. Alam, M.M.R. Howlader, N.-X. Hu, M.J. Deen, Morphology and electrical properties of inkjet-printed palladium/palladium oxide, *J. Mater. Chem. C* 5 (2017) 1893–1902, <http://dx.doi.org/10.1039/C6TC05196K>.
- [25] H. Lee, T.K. Choi, Y.B. Lee, H.R. Cho, R. Ghaffari, L. Wang, et al., A graphene-based electrochemical device with thermoresponsive microneedles for diabetes monitoring and therapy, *Nat. Nanotechnol.* 11 (2016) 566–572, <http://dx.doi.org/10.1038/nnano.2016.38>.
- [26] W. Honda, S. Harada, T. Arie, S. Akita, K. Takei, Wearable, human-interactive, health-monitoring, wireless devices fabricated by macroscale printing techniques, *Adv. Funct. Mater.* 24 (2014) 3299–3304, <http://dx.doi.org/10.1002/adfm.201303874>.
- [27] H.Y.Y. Nyein, W. Gao, Z. Shahpar, S. Emaminejad, S. Challa, K. Chen, et al., A wearable electrochemical platform for noninvasive simultaneous monitoring of Ca²⁺ and pH, *ACS Nano* 10 (2016) 7216–7224, <http://dx.doi.org/10.1021/acsnano.6b04005>.
- [28] W. Gao, S. Emaminejad, H.Y.Y. Nyein, S. Challa, K. Chen, A. Peck, et al., Fully integrated wearable sensor arrays for multiplexed in situ perspiration analysis, *Nature* 529 (2016) 509–514, <http://dx.doi.org/10.1038/nature16521>.
- [29] Z. Xu, Q. Dong, B. Otieno, Y. Liu, I. Williams, D. Cai, et al., Real-time in situ sensing of multiple water quality related parameters using micro-electrode array (MEA) fabricated by inkjet-printing technology (IPT), *Sens. Actuator B Chem.* 237 (2016) 1108–1119, <http://dx.doi.org/10.1016/j.snb.2016.09.040>.
- [30] S. Mross, T. Zimmermann, N. Winkin, M. Kraft, H. Vogt, Integrated multi-sensor system for parallel in-situ monitoring of cell nutrients, metabolites, cell density and pH in biotechnological processes, *Sens. Actuator B Chem.* 236 (2016) 937–946, <http://dx.doi.org/10.1016/j.snb.2016.03.086>.
- [31] D. Kirsanov, A. Korepanov, D. Dorovenko, E. Legin, A. Legin, Indirect monitoring of protein A biosynthesis in E.coli using potentiometric multisensor system, *Sens. Actuator B Chem.* 238 (2017) 1159–1164, <http://dx.doi.org/10.1016/j.snb.2016.02.073>.
- [32] M.H. Banna, H. Najjaran, R. Sadiq, S.A. Imran, M.J. Rodriguez, M. Hoorfar, Miniaturized water quality monitoring pH and conductivity sensors, *Sens. Actuator B Chem.* 193 (2014) 434–441, <http://dx.doi.org/10.1016/j.snb.2013.12.002>.
- [33] B. Zhou, C. Bian, J. Tong, S. Xia, Fabrication of a miniature multi-parameter sensor chip for water quality assessment, *Sensors* 17 (2017) 157, <http://dx.doi.org/10.3390/s17010157>.
- [34] S. Pan, M.J. Deen, R. Ghosh, Low-cost graphite-based free chlorine sensor, *Anal. Chem.* 87 (2015) 10734–10737, <http://dx.doi.org/10.1021/acs.analchem.5b03164>.
- [35] Y. Qin, A.U. Alam, S. Pan, M.M.R. Howlader, R. Ghosh, P.R. Selvaganapathy, et al., Low-temperature solution processing of palladium/palladium oxide films and their pH sensing performance, *Talanta* 146 (2016) 517–524, <http://dx.doi.org/10.1016/j.talanta.2015.08.062>.
- [36] J. Orozco, A. Baldi, R. Baena, A. Cadarso, A. Bratov, C. Jimenez, Portable system based on microsystems for environmental monitoring applications, *Meas. Sci. Technol.* 18 (2007) 935–940, <http://dx.doi.org/10.1088/0957-0233/18/3/048>.
- [37] (a) A. Santos, M.J. Deen, L.F. Marsal, Low-cost fabrication technologies for nanostructures: state-of-the-art and potential, *Nanotechnology* 26 (2015) 42001, <http://dx.doi.org/10.1088/0957-4484/26/4/042001>; (b) M.W. Shinwari, D. Zhitomirsky, I.A. Deen, P.R. Selvaganapathy, M.J. Deen, Dolf Landheer, Microfabricated reference electrodes and their biosensing applications, *Sensors* 10 (2010) 1679–1715, <http://dx.doi.org/10.3390/s100301679>.
- [38] R. Olivé-Monllau, J. Orozco, C. Fernández-Sánchez, M. Baeza, J. Bartrolí, C. Jimenez-Jorquera, et al., Flow injection analysis system based on amperometric thin-film transducers for free chlorine detection in swimming pool waters, *Talanta* 77 (2009) 1739–1744, <http://dx.doi.org/10.1016/j.talanta.2008.10.015>.
- [39] Y. Qin, M.M.R. Howlader, M.J. Deen, Y.M. Haddara, P.R. Selvaganapathy, Polymer integration for packaging of implantable sensors, *Sens. Actuator B Chem.* 202 (2014) 758–778, <http://dx.doi.org/10.1016/j.snb.2014.05.063>.

Biographies



Yiheng Qin has completed his Ph.D. degree in Department of Electrical and Computer Engineering at McMaster University, Hamilton, Canada. He received his B.S. degree in Electronics Science and Technology (Microelectronics Technology) from Southwest Jiaotong University, Chengdu, China, in 2009, and received his M.S. degree in Microtechnology from Chalmers University of Technology, Göteborg, Sweden, in 2011. His research interests include sensors, printed electronics, nano-integration technologies, with a focus on the use of polymeric materials for biomedical and environmental applications.



Arif Ul Alam is currently pursuing a Ph.D. degree in Department of Electrical and Computer Engineering at McMaster University, Hamilton, Canada. He received a B.S. degree in Electrical and Electronic Engineering from Islamic University of Technology, in Dhaka, Bangladesh in 2008 and received a M.A.Sc. degree in Electrical and Computer Engineering at McMaster University, Hamilton, ON, Canada in 2013. His research focuses on design and development of low cost electrochemical sensors for environmental and health applications.



Si Pan is a Ph.D. student in the Department of Chemical Engineering at McMaster University, Hamilton, Canada. He received a B.E. in 2010 in Polymer Materials and Engineering in the College of Chemistry at Jilin University, Changchun, China and an M.A.Sc. in 2013 in Bioseparation in the Department of Chemical Engineering at McMaster University. He is currently researching free chlorine sensors for drinking water quality monitoring.



Shurong Dong received the B.S. and Ph.D. degrees in electronics engineering from Zhejiang University, Hangzhou, China, in 1994 and 2003, respectively. He is currently a Professor and the Deputy Director of Microelectronics and Photo-Electronics Institute with Zhejiang University. His current research interests include flexible electronics, biosensors, and microelectronics reliability.

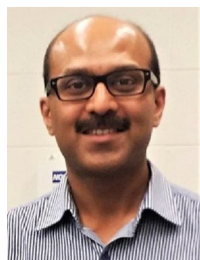


Matiar M. R. Howlader received a B. Sc. Eng. degree in Electrical and Electronic Engineering from Khulna University of Engineering and Technology, in Khulna, Bangladesh, in 1988, and his M.S. and Ph.D. degrees in Nuclear Engineering from Kyushu University, Fukuoka, Japan, in 1996 and 1999, respectively. Currently, Dr. Howlader is an Assistant Professor in the Department of Electrical and Computer Engineering at McMaster University, Hamilton, ON, Canada. His research focuses, broadly, on the design, fabrication, characterization, and development of new materials and novel engineered components to create integrated sensing systems for environmental and health applications. For this, in particular, he develops

foundational technologies in the surface activated nanobonding currently, emphasizing on the flexible, soft and hard materials integration for the sensing systems that enable energy harvesting.



Chih-Hung Chen received the B.S. degree in electrical engineering from National Central University, Chungli, Taiwan in 1991, the M.S. degree in engineering science from Simon Fraser University, Burnaby, BC, Canada in 1997 and the Ph.D. degree in the Department of Electrical and Computer Engineering, McMaster University, Hamilton, ON, Canada in 2002. He is currently an Associate Professor in the Department of Electrical and Computer Engineering, McMaster University, Hamilton, ON, Canada. His research interest is in the area of noise characterization and modeling of nano-scale semiconductor devices, and the design of low-noise, low-power integrated circuits (IC) and sensor interface electronics.



Raja Ghosh is Professor of Chemical Engineering at McMaster University, Canada. His research interests include purification of biopharmaceuticals using membrane technology and chromatography, water analysis and treatment, and membrane bioreactors for biopharmaceutical production. He has authored two books, *Protein Bioseparation Using Ultrafiltration* and *Principles of Bioseparations Engineering*, has published over 100 research papers in peer-reviewed journals, and several chapters in edited books. He serves on the editorial board of *Membrane Water Treatment, Processes*, *Journal of Pharmaceutical Analysis and Recent Patents in Biotechnology*. Professor Ghosh obtained his Doctorate degree in Engineering from Oxford University, United Kingdom. He began his academic career as a faculty member at Oxford University before moving to McMaster University.



M. Jamal Deen is currently Distinguished University Professor and Senior Canada Research Chair in Information Technology, McMaster University. His current research interests are nanoelectronics, optoelectronics, nanotechnology and their emerging applications to health and environmental sciences. Dr. Deen's research record includes more than 540 peer-reviewed articles, two textbooks on "Silicon Photonics- Fundamentals and Devices" and "Fiber Optic Communications: Fundamentals and Applications", 6 awarded patents that have been used in industry, and 17 best paper/poster/presentation awards. His other awards and honors include the Callinan Award as well as the Electronics and Photonics Award from the Electrochemical Society; a Humboldt Research Award from the Alexander von Humboldt Foundation; the Eadie Medal from the Royal Society of Canada; McNaughton Gold Medal (highest award for engineers), the Fessenden Medal and the Ham Education Medal, all from IEEE Canada. In addition, he was awarded the four honorary doctorate degrees in recognition of his exceptional research and scholarly accomplishments, professionalism and service. Dr. Deen has also been elected Fellow status in ten national academies and professional societies including The Royal Society of Canada – The Academies of Arts, Humanities and Sciences, the IEEE, the American Physical Society and the Electrochemical Society. Currently, he is serving as the elected President of the Academy of Science, The Royal Society of Canada.



Nan-Xing Hu is a Research Fellow with the Xerox Research Centre of Canada (XRCC), which is specialized in advanced materials research and development. He joined Xerox in 1994 and currently oversees the R&D activities in the Xerographic Consumables Program. Nan-Xing also supervises Ph.D. students from McMaster University who are conducting materials research for printed electronics applications. His research interests focus on organic electronic materials, nano-materials/structures, and their applications. Nan-Xing received his Ph.D. in organic chemistry from Hiroshima University (Japan) in 1989.



Hao Jin received his B.S. and Ph.D. degrees in electronic science and technology from Zhejiang University, P.R. China, in 2001 and 2006, respectively. He is currently an associate professor in the college of information science & electronic engineering at Zhejiang University, P.R. China. His research interests include sensors for health-care and environments application, vacuum science and technology, micro/nano piezoelectric devices, RF MEMS, and flexible electronics.